Limits to Arbitrage in the Indian Equity Market

BS Project (EC 495) Presentation

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November 2026



Outline

1. Introduction
2. Background and Literature
3. Institutional Setting
4. Theoretical Model
5. Data and Methodology
6. Results I: The F&O Ban
7. Results II: Costs, Capital and Position Limits
8. Results III: Short-Sale Constraints
9. Conclusions and Future Scope

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Motivation

- A share and a future on it give the same payoff, so their prices should be consistent; arbitrage closes any gap.
- In practice arbitrage is limited: it needs capital, carries risk, costs money and often needs a short position (Shleifer and Vishny, 1997; Gromb and Vayanos, 2010).
- India changes almost every such limit on a known date:
  - the securities transaction tax (STT) on futures rose in 2023, 2024 and 2026;
  - only F&O stocks can be shorted, and stocks enter and leave that segment;
  - arbitrage-fund assets quadrupled after the 2023 change in debt-fund taxation;
  - position limits are set by formula and changed in October 2025.
- India also has an unusual rule, the F&O ban, which no study has examined.

The F&O ban

- Market-wide open interest above **95%** of the market-wide position limit (MWPL): ban from the next day
- **Everyone**, not only arbitrageurs, may only reduce positions
- Ban ends when open interest falls to **80%**
- 1,788 spells in 141 stocks, 2015 to 2026

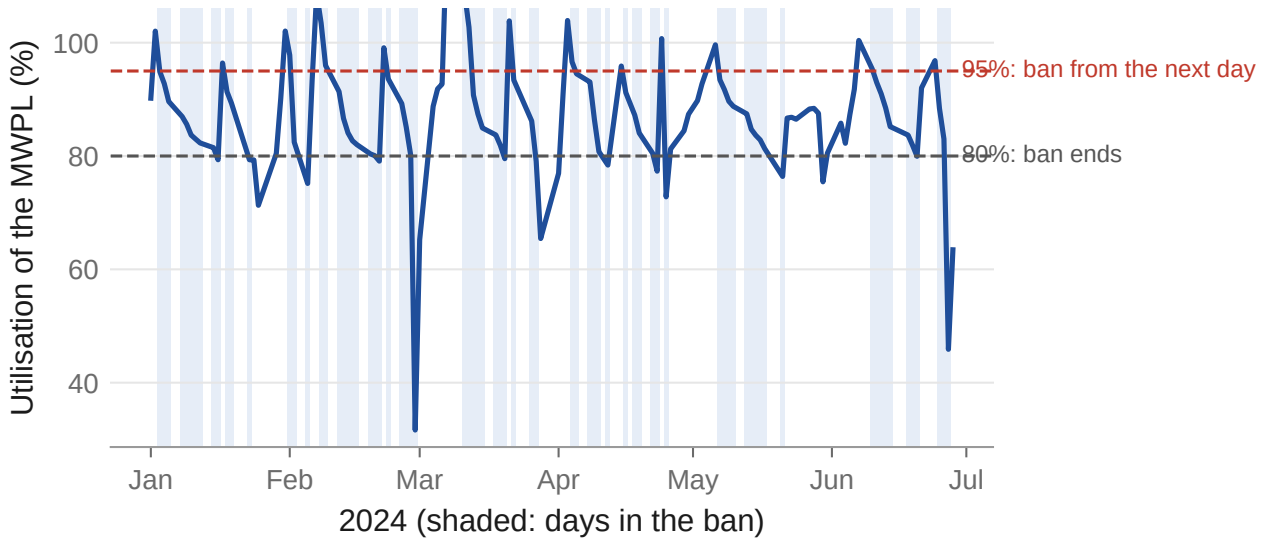


Figure 1: Utilisation of the MWPL for SAIL, January to June 2024; shaded: ban days.

Research question and objectives

Research question

How strongly do limits to arbitrage bind in the Indian equity market, which of them matter, and how much mispricing do they explain?

1. Build one model in which every Indian limit is a parameter, and derive testable predictions with falsifiers.
2. Freeze the tests before relating any outcome to any treatment.
3. Measure mispricing (the futures-cash basis, earnings drift, price delay) from public exchange data.
4. Test how bans, costs, capital, position limits and short-sale access shape mispricing.
5. Draw out what the results imply for the regulation of position limits and short selling.

Research gaps

Table 1: Earlier Indian estimates of the futures-cash band

Study	Sample	Data	Band
Aggarwal (2015)	141 stocks, 2013	One-second	-15 to $+3$ bp
Shankar, Sankar and Kiran Kumar (2019)	102 stocks, 2007 to 2009	Five-minute	-39 to -10 bp
Slivka, Shah and Wu (2011)	Cost calculation	Fee schedules	Twice mature markets
This study	**360 stocks, 2015 to 2026**	**Daily**	$-$**34.5** to $+$**23.9** bp

- Earlier work measures intraday bands over short samples and never uses the ban.
- No study has estimated the effects of a reduce-only rule that binds every participant.
- Limits are usually studied one at a time, not side by side within one market.

Main contributions

1. The first study of India's F&O ban, modelled as a reduce-only rule for every participant.
2. A model showing that the ban makes correction one-sided and its effect on the level of the basis ambiguous in sign; and that the position limit is not a valid instrument for ban status.
3. Pre-registered tests with falsifiers, frozen with a time stamp; all results reported.
4. Estimates of the no-arbitrage band over a long daily panel, related to dated changes in costs, capital and position limits.
5. A constraint index for every listed Indian stock, released as a data set.

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Why arbitrage is limited

- **Capital:** arbitrageurs invest other people's money; losses bring withdrawals when gaps widen (Shleifer and Vishny, 1997).
- **Risk:** noise-trader risk deters short-horizon arbitrageurs (De Long et al., 1990); the result needs market-wide noise and is fragile (Loewenstein and Willard, 2006).
- **Costs and short sales:** a shorting cost makes the band one-sided (Gromb and Vayanos, 2010).
- **Short-sale constraints and prices:** overpricing (Miller, 1977); slower incorporation of bad news (Diamond and Verrecchia, 1987).
- **The band:** threshold and smooth-transition models of futures and cash (Dwyer, Locke and Yu, 1996; Tse, 2001; Ters and Urban, 2020).

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Dated changes to the limits, 2015 to 2026

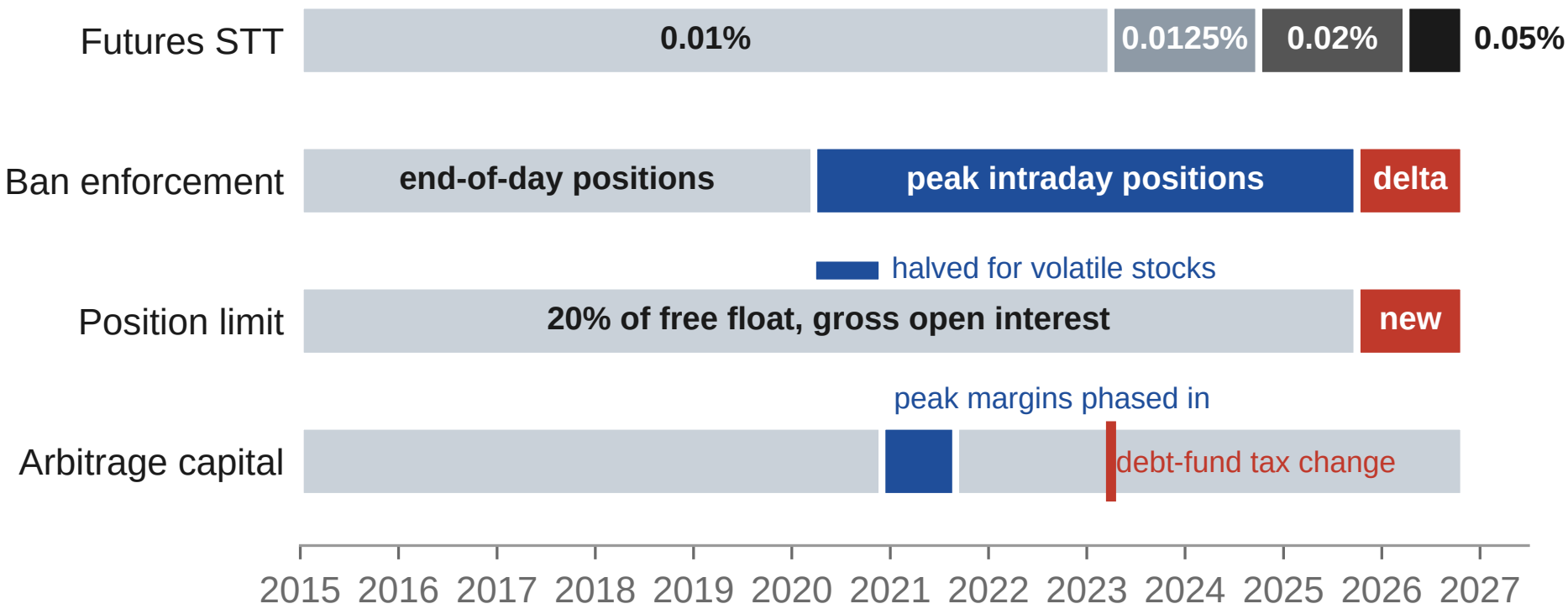


Figure 2: Futures STT, enforcement of the ban, the position-limit rule and arbitrage capital.

- To 22 March 2020 ban violations were judged on end-of-day positions; then on peak intraday positions; from October 2025 on end-of-day delta.

Position limits and short selling

Table 2: The market-wide position limit over the sample

Period	MWPL	Open interest tested
To 30 Sep 2025	20% of free float	Gross, futures and options
Mar to Nov 2020	Halved for volatile or heavily used stocks	Gross
From 1 Oct 2025	Lower of 15% of free float and 65 times average daily delivery; floor 10%	Delta-adjusted

- Only F&O stocks are eligible for short selling; naked shorting is barred; securities lending is thin.
- The practical overnight short is a single-stock future, so F&O entry and exit change a stock's short-sale constraint at once.

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Set-up

- Basis $m = \ln F - \ln F^*$, with $F^* = (S - \mathrm{PV}(D))\, e^{r(T-t)}$.
- Outside traders create demand pressure u toward futures; an arbitrage sector holds a units of cash-and-carry. Market clearing:

$$m = \frac{u - a}{\Phi}.$$

- The sector maximises

$$\Pi(a) = a m - \tfrac{\gamma}{2} a^2 - c_+ a^+ - c_- a^-, \qquad \gamma = \alpha \sigma^2,$$

  where c_+ is the cost of cash-and-carry and c_- the cost of reverse arbitrage.
- Every Indian limit is a parameter: costs ($c_\pm$), short sales (c_-), capital and risk (γ), position limits (caps on a).

The no-arbitrage band

Proposition 1 (Band)

Arbitrage starts only when m leaves $[-c_-, c_+]$. Inside, $\partial m/\partial u = 1/\Phi$; outside, $\partial m/\partial u = \gamma/(\gamma\Phi + 1)$.

- Costs set the edges one for one
- Capital and risk set how far m travels beyond them
- $c_- > c_+$ in India: the band sits below zero

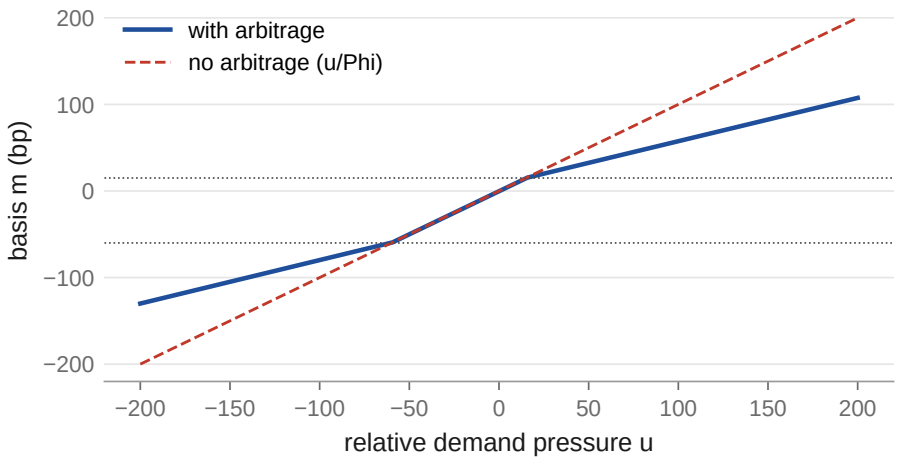


Figure 3: Equilibrium basis against demand pressure (illustrative costs).

The F&O ban as a reduce-only rule

Assumption 1 (Ban)

(B1) Arbitrageurs may only reduce: $0 \leq a_t \leq a_{t-1}$. *(B2) New futures demand is blocked; a share* χ *moves to the stock, the rest is withdrawn.*

Proposition 2 (One-sided correction)

Rich futures are no longer corrected; cheap futures are, at a floor $\gamma a_t - c_u$ *that wears out as positions are unwound.*

Proposition 3 (Level effect)

The ban raises the basis when pressure comes from closing trades and lowers it when new demand keeps arriving: its sign is set by the composition of order flow.

Further results: trading in a ban ($\Delta OI = -V$), anticipation below 95%, the MWPL is not an instrument for bans, bad news is incorporated more slowly, Miller overpricing.

A simulated ban

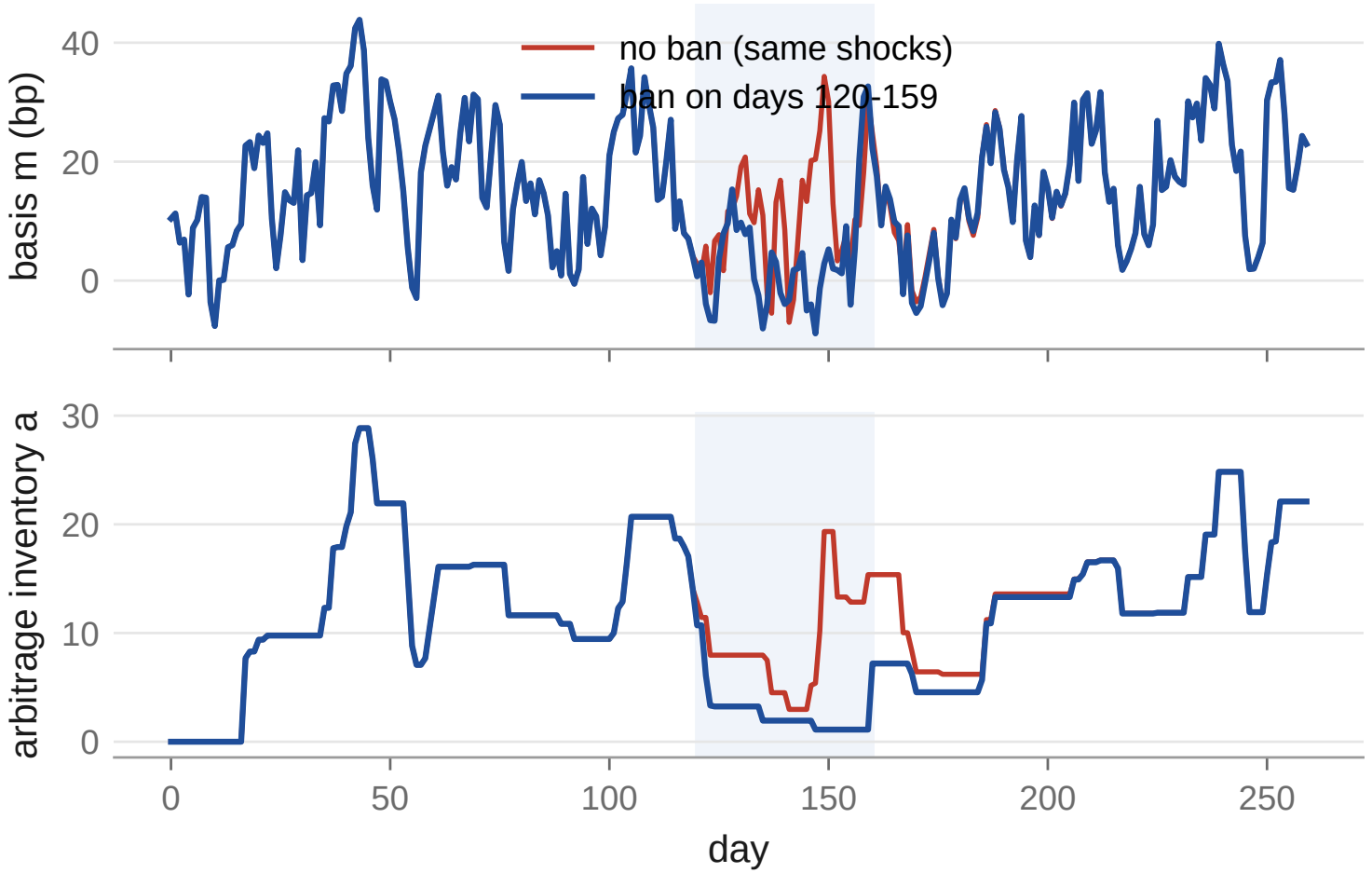


Figure 4: Same shocks with and without a ban on days 120 to 159: basis (top) and arbitrage inventory (bottom). Illustrative parameters.

Hypotheses and falsifiers

Table 3: Pre-specified hypotheses; the first four form the primary family (Holm correction)

	Prediction	Contradicted if
H8	In a ban, futures volume falls and every trade closes positions	Futures activity unchanged
H7	In a ban, rich futures revert more slowly	No asymmetry, or the opposite
H3	Band edges move with the STT	Edges unrelated to costs
H6	F&O entry lowers price delay and bad-news drift	No change, or the opposite
Level	The level effect follows the composition of open interest	Sign unrelated to composition
H9	The upper tail grows as utilisation nears 95%	Jump only at 95%
H2, H1	Bad news slower in bans; wider bands with frictions	Symmetric; no relation
H4, H5	Scarce capital widens the band; one-sided anomalies	No relation; symmetric

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Data

Table 4: Public data sources

Source	Content	Coverage
NSE open interest, ban lists	MWPL, open interest, ban status	2010 to 2026
NSE bhavcopies, delivery files	Futures and cash prices, volumes, delivery	2015 to 2026
NSE corporate events	Dividends, splits, results dates	2014 to 2026
NSE securities lending	Daily lending trades and fees	2015 to 2026
AMFI	Market capitalisation; arbitrage-fund flows	2017 to 2026

Checks

Next-day flags match the official ban list for 99.9% of flags; results are dated from the exchange's first dissemination time; a full rerun reproduces every number.

The futures-cash basis

$$m_{it} = \ln \bar{F}_{it} - \ln\left(\bar{S}_{it} - \mathrm{PV}_{it}(D)\right) - r_t \tau_{it}$$

- Volume-weighted prices, not settlement prices (set at cost of carry on thin days)
- Near-month contract, at least 5 contracts traded, no dividend in its life
- 486,040 stock-days, 371 stocks

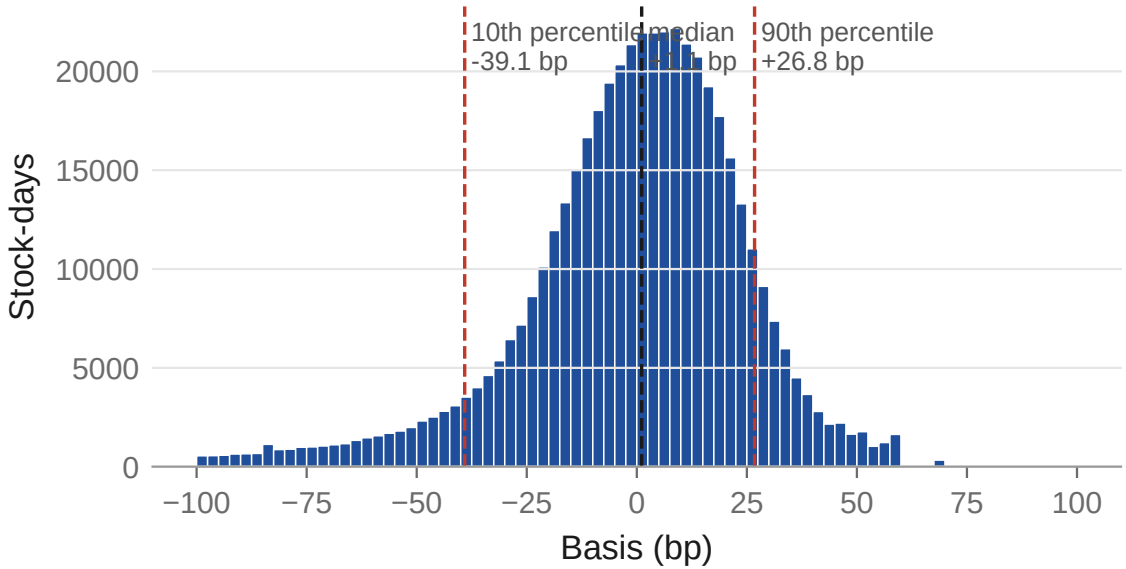
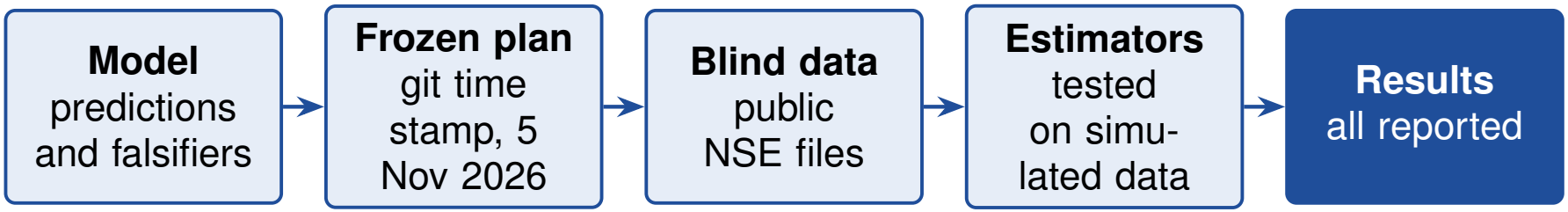


Figure 5: Distribution of the basis (bp).

Research design



- **Estimators:** two-way fixed effects with clustered errors; threshold and smooth-transition models; matched event studies; density test and regression discontinuity; Callaway and Sant'Anna difference-in-differences.
- **Inference:** primary family Holm-corrected at 5%; secondary tests unadjusted and read as a pattern.

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Trading in a ban (H8)

Table 5: Ban days against the same stock's non-ban days at 80 to 95% utilisation

Outcome	Effect (s.e.)	Non-ban mean
log(1 + futures contracts)	−2.00 (0.11)	
log(1 + cash shares)	+0.073 (0.027)	
Delivery share	+0.041 (0.005)	0.26

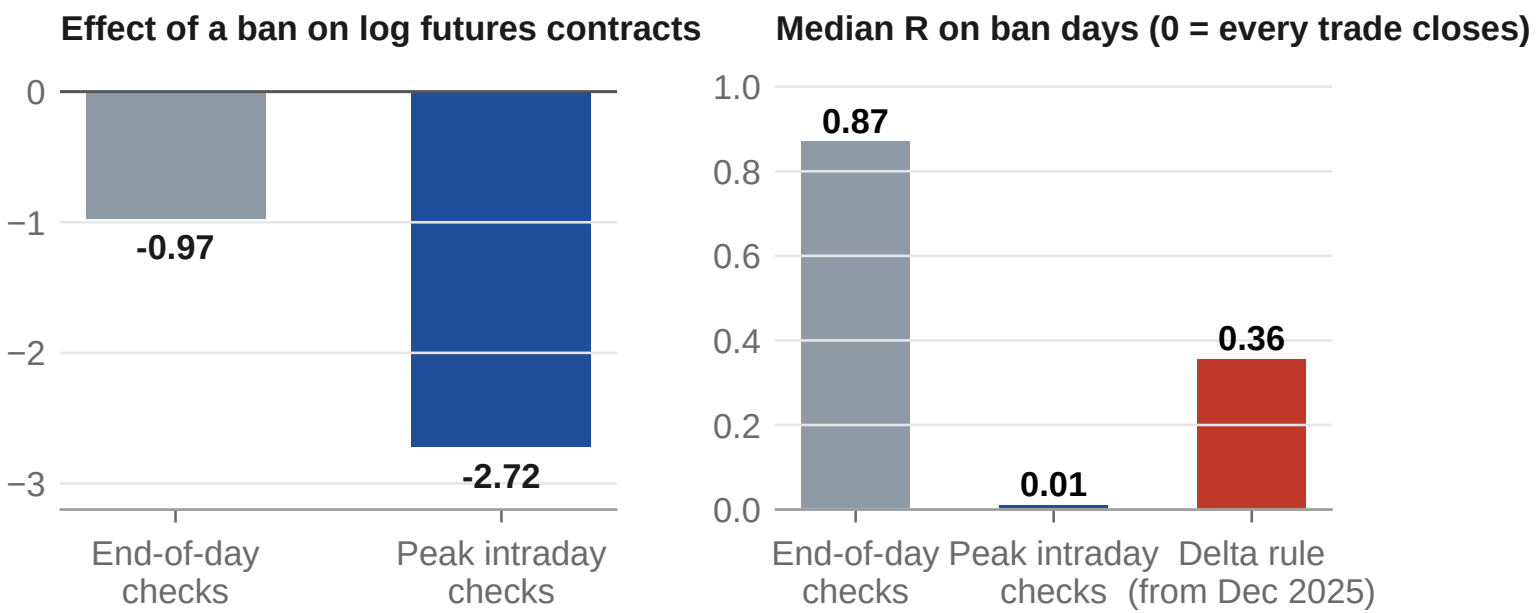


Figure 6: By enforcement regime: futures effect (left) and median $R = (\Delta OI + V)/V$ on ban days (right).

Enforcement decides whether the rule binds

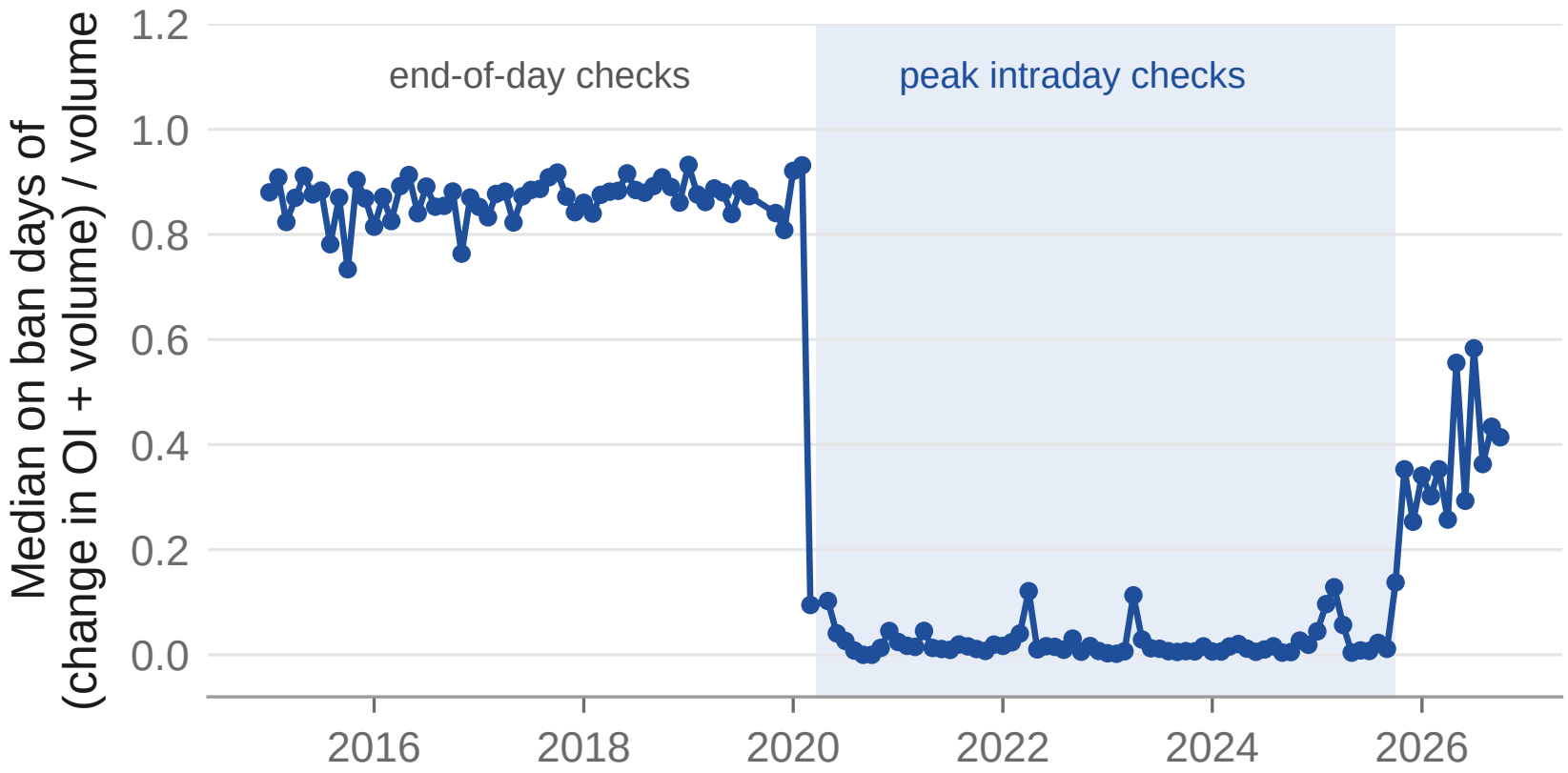


Figure 7: Monthly median of R on ban days; $R = 0$ when every trade closes a long and a short.

Under intraday checks (March 2020 to September 2025) the median R is 0.011: every trade in a ban closed positions.

Does correction become one-sided? (H7)

Table 6: Reversion slopes

	Normal	Change in ban
Upper	−0.712	**−0.089**
Middle	−0.393	+0.099
Lower	−0.204	**+0.081**

Prediction rejected

Rich futures revert *faster* in a ban, mostly in the first five days of a spell.

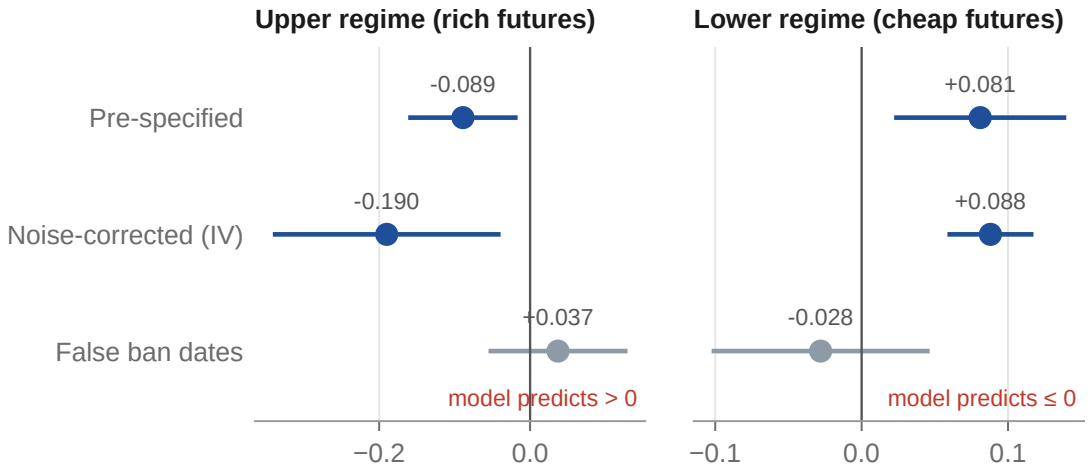


Figure 8: Change in slope: pre-specified, noise-corrected, false ban dates.

The level of the basis when a ban starts

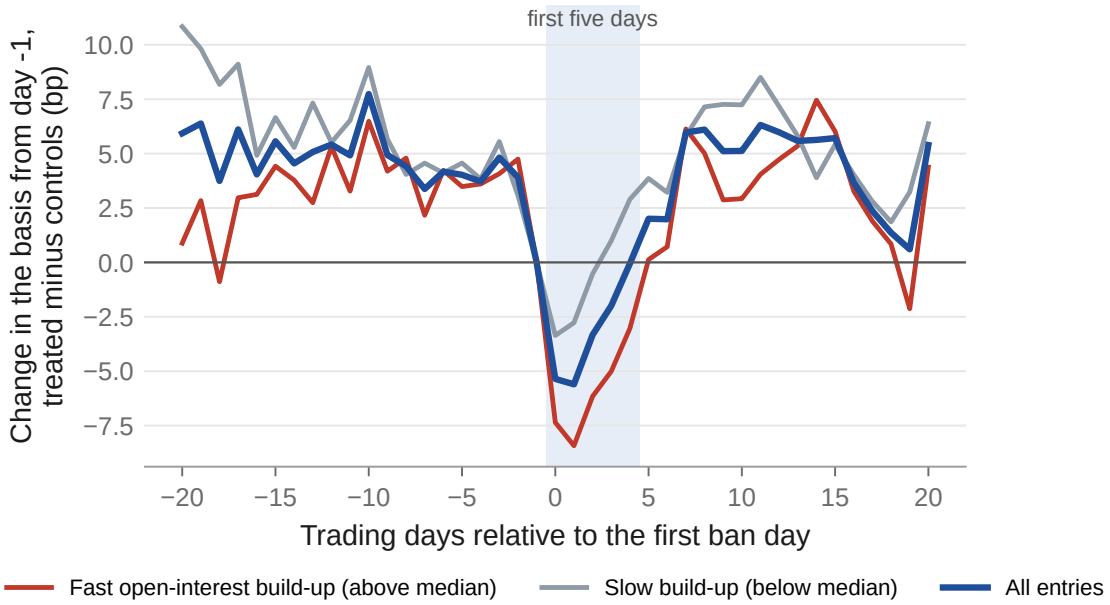


Figure 9: Treated minus matched controls, by event day (889 entries).

- Basis falls by 6.3 bp in the first days; returns after about a week
- Rising open interest before entry: -6.8 bp; falling: $+8.4$ bp (30 events)
- Difference -15.2 bp ($p = 0.083$); median split $p = 0.025$
- Same for long and short build-ups; nothing at false entries

Consistent with blocked new futures demand (a reading formed after the results).

The 95% trigger: no bunching

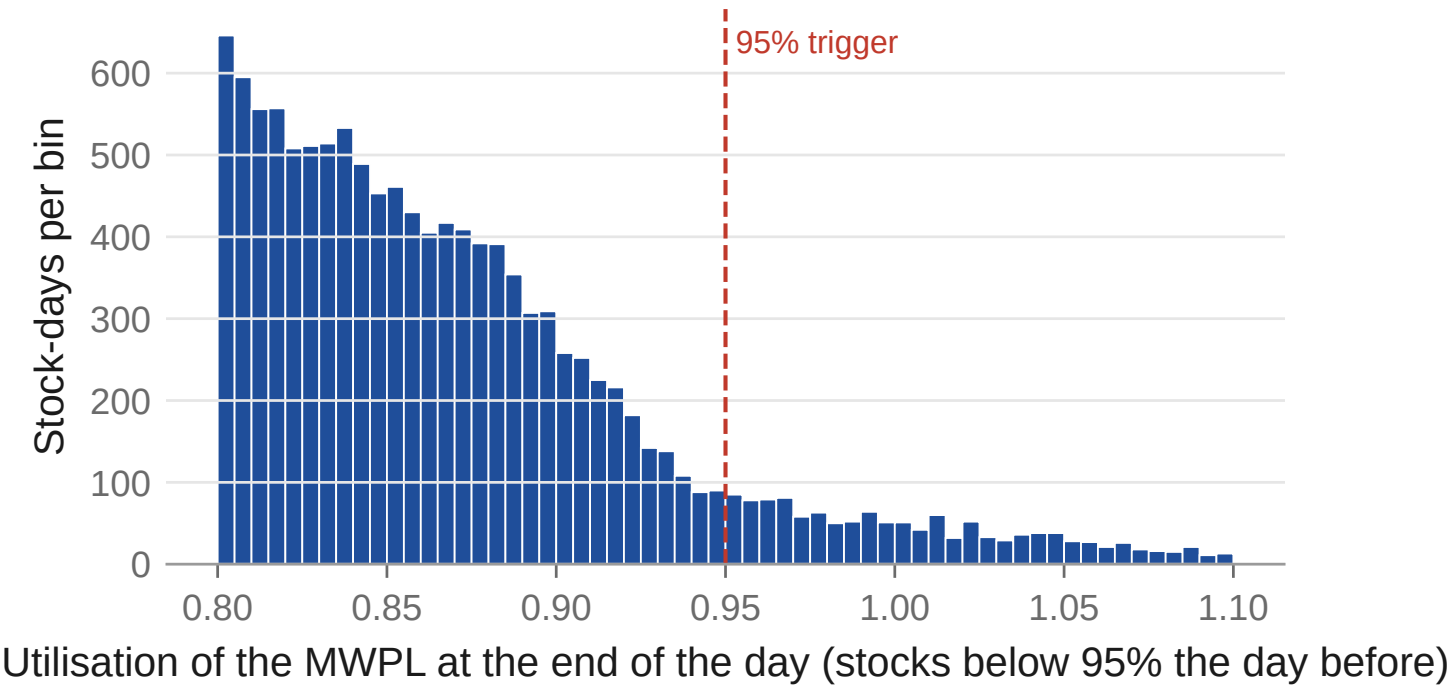


Figure 10: End-of-day utilisation of stocks that were below 95% the day before, 2015 to September 2025.

- Cattaneo, Jansson and Ma test: $T = -0.18$, $p = 0.86$; robust to bandwidth, order and enforcement period.
- Utilisation accelerates rather than stalls near the trigger: traders do not hold back.

The 95% trigger: regression discontinuity

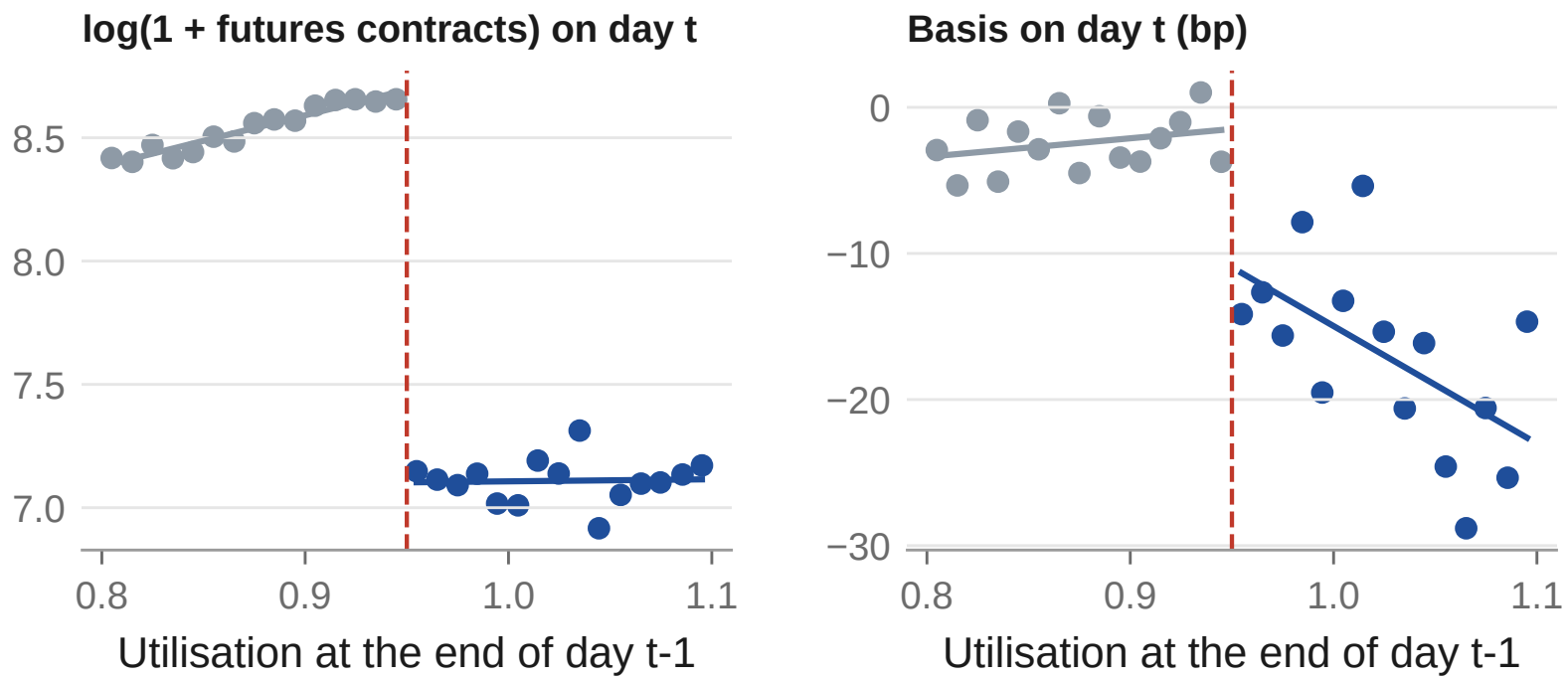


Figure 11: Binned means by utilisation at the end of the previous day, stocks not in a ban the day before.

- Futures trading drops by 1.53 log points at 95%; R falls by 0.66; delivery share rises.
- The basis is 12.9 bp lower above the cutoff, but already 11.3 bp lower on the crossing day.
- Hysteresis (80 to 95%): being in a ban adds $+0.4$ bp ($p = 0.64$).

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The no-arbitrage band

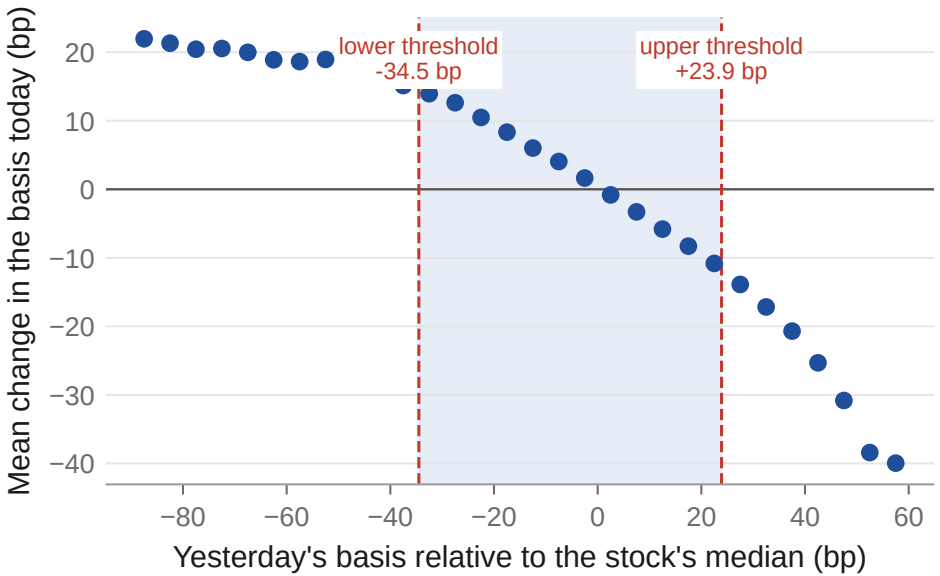


Figure 12: Mean daily change by yesterday's deviation.

Table 7: Pooled threshold model

	Lower	Middle	Upper
Threshold	−34.5		+23.9
Slope	−0.084	−0.468	−0.809
IV slope	−0.121	−0.192	−0.141

Linearity rejected (sup-F 7,471). Instrumenting shows most daily reversion is price noise.

Costs and the band (H3)

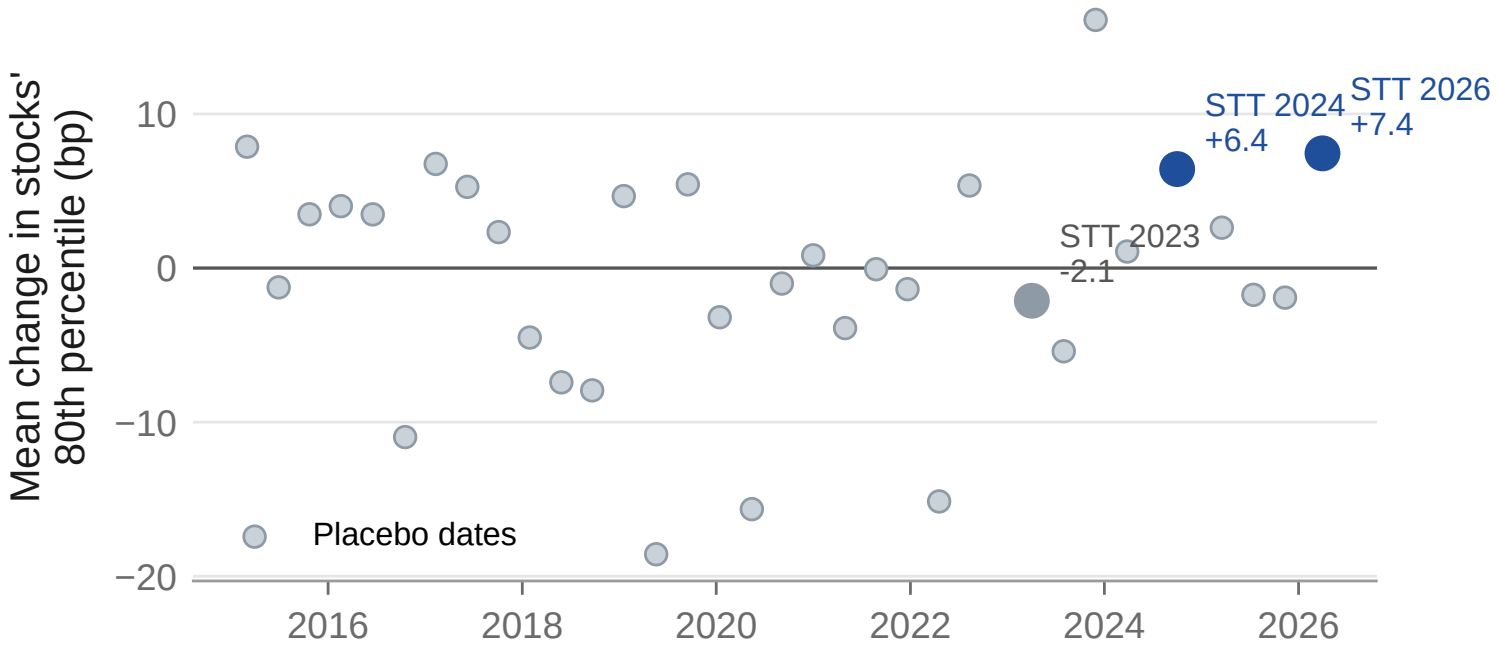


Figure 13: Mean change in stocks' 80th percentile of the basis at the STT dates and at 30 placebo dates (bp).

- Pre-specified: $+6.96$ bp (s.e. 0.61) pooling 2024 and 2026; ordering 2026 $>$ 2024 $>$ 2023 holds.
- But the placebo s.d. is 7.4 bp: the rise is mostly a common shift, larger than the cost changes.

The cost of a new cash-and-carry position

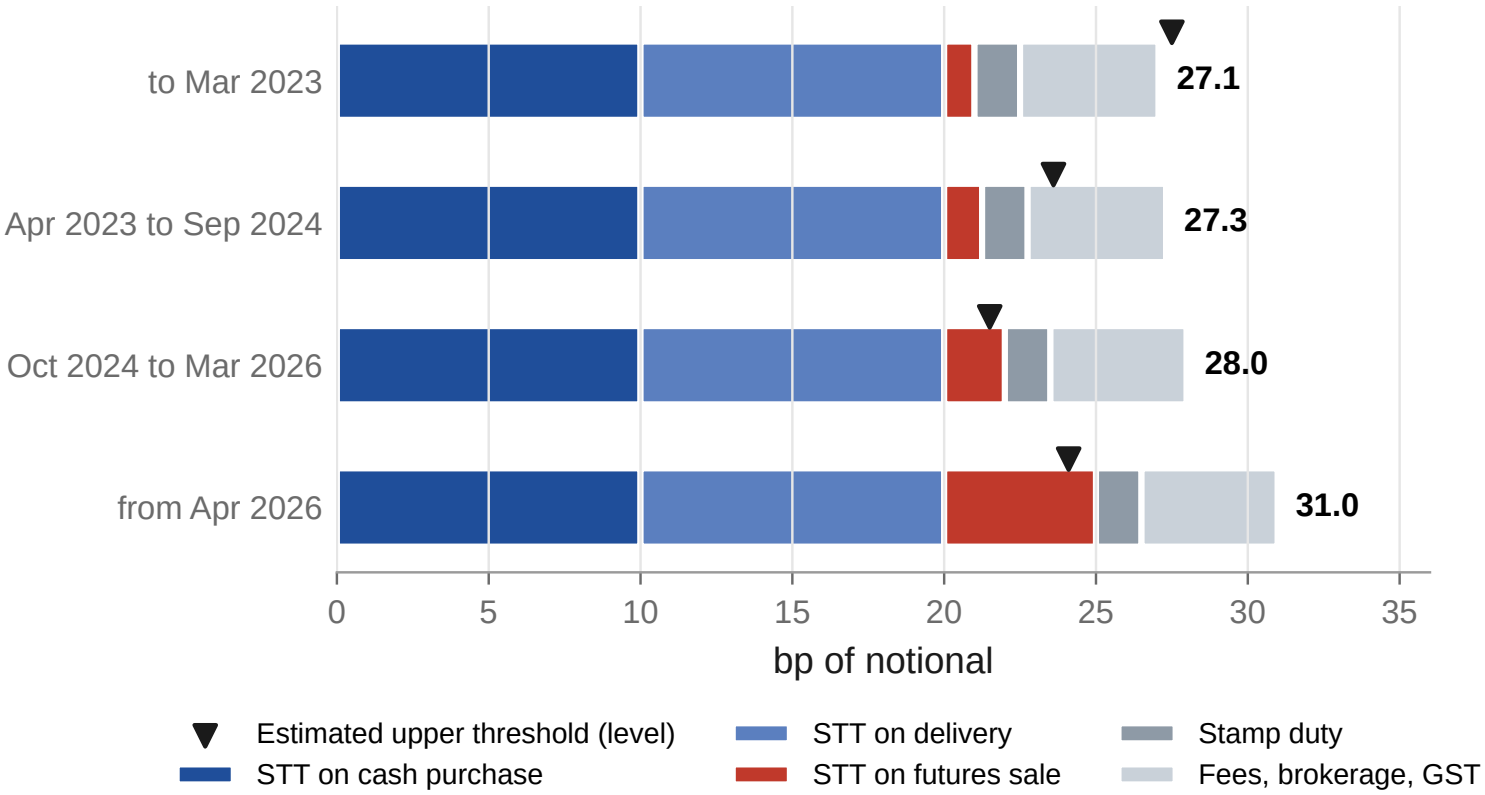


Figure 14: Round trip of an overnight position held to delivery, by STT period (bp of notional).

- A new round trip costs 27 to 31 bp, above the estimated upper edge; rolling costs only 4 to 8 bp.
- New capital enters only at large gaps; funds that roll lean against small ones.

Arbitrage capital (H4)

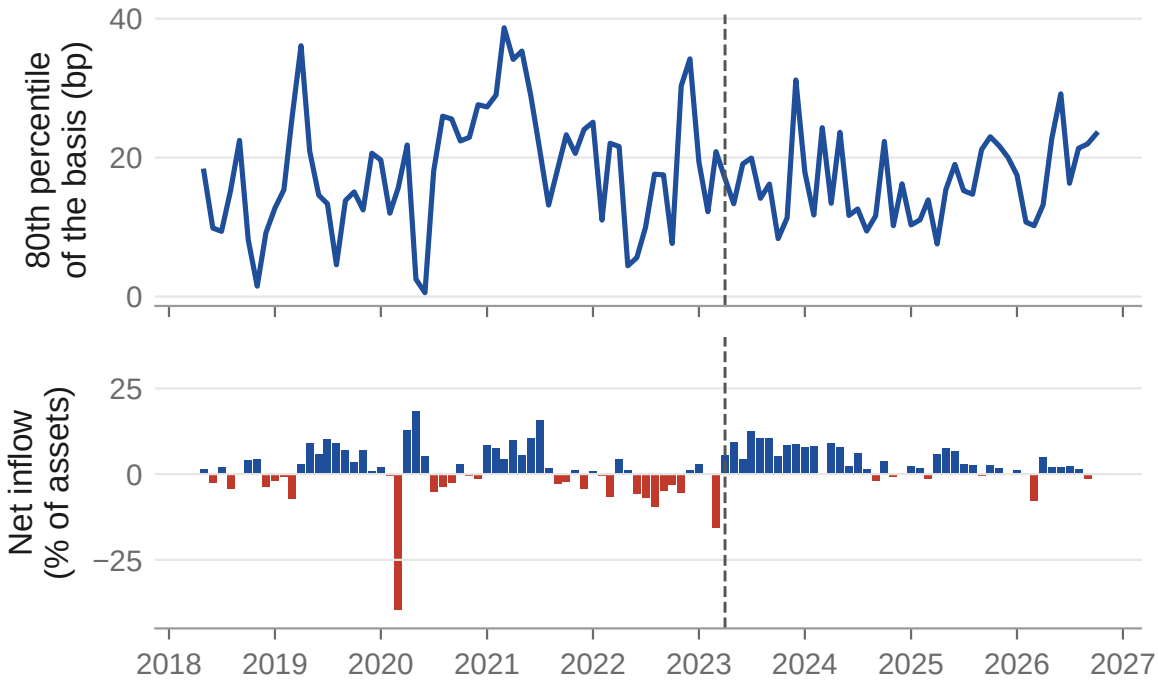


Figure 15: Monthly 80th percentile of the basis (top) and net inflow into arbitrage funds (bottom).

- -17.5 bp per unit of lagged flow ($p = 0.11$); clearer in first differences (-20.6, s.e. 6.8).
- Visible before April 2023, not after, when assets grew fastest.

Anticipation below the trigger (H9)

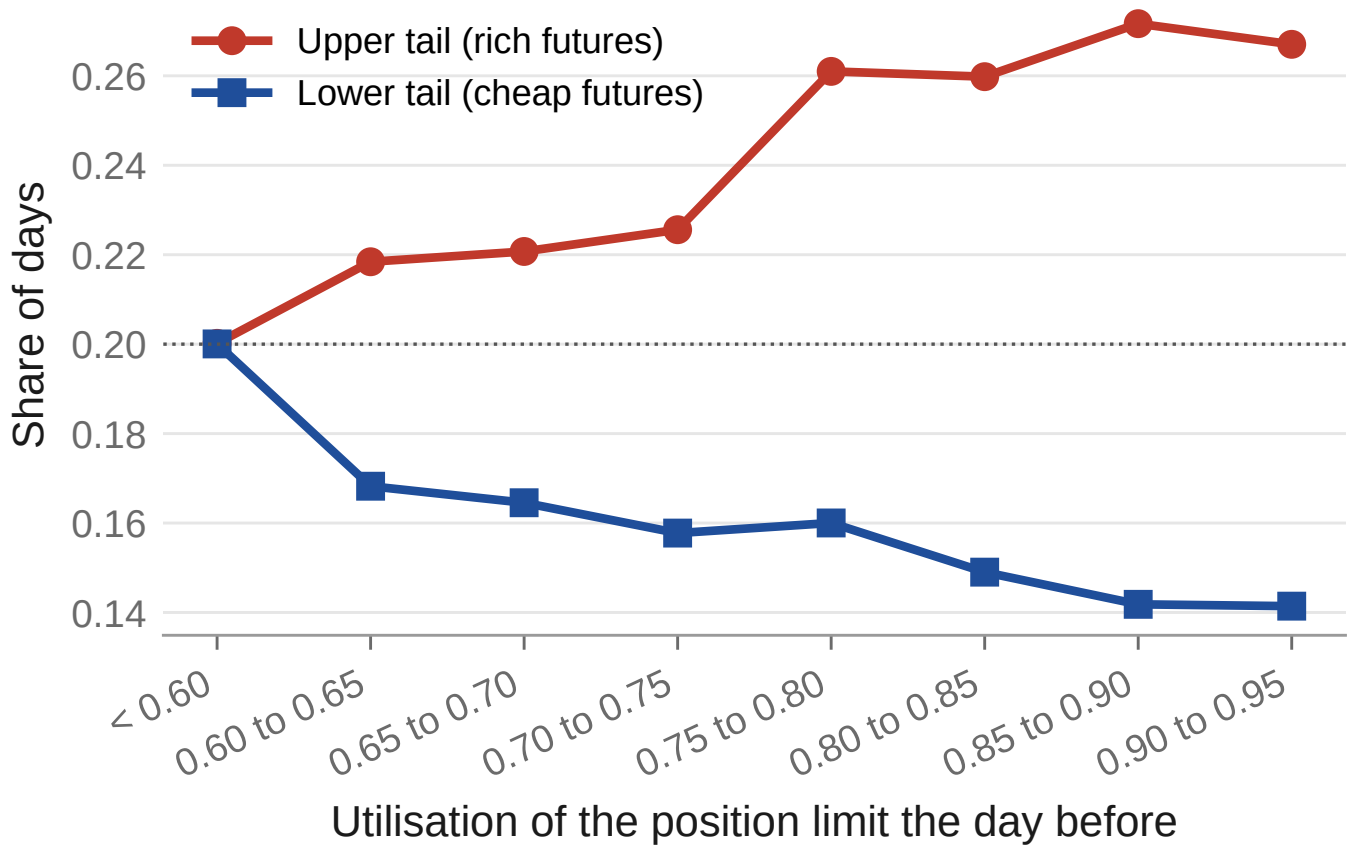


Figure 16: Share of non-ban days in the stock's upper and lower tail, by utilisation the day before.

- Upper-tail share rises from 0.20 to 0.27 toward 95% ($p = 0.019$); the lower tail is flat with date effects.
- October 2025 reform: right direction, not significant ($p = 0.11$).

Restoring the position limit in 2020

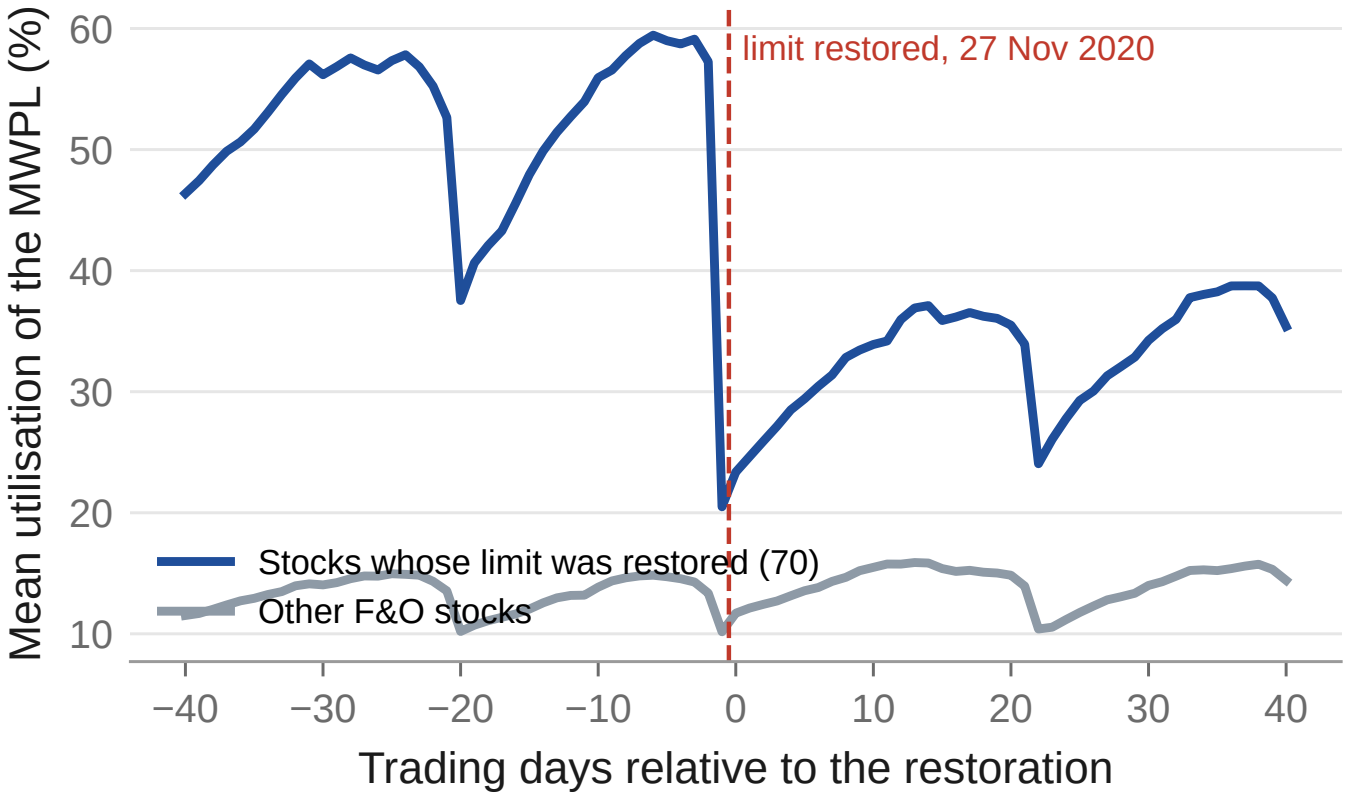


Figure 17: Mean utilisation of the 70 stocks whose halved limit was restored on 27 November 2020, and of others.

- Utilisation −0.21, ban days −4.6 points, upper tail −6.8 points; level unchanged.
- A looser limit made rich futures less common, as the anticipation result predicts.

The 2025 rules and “spurious” bans

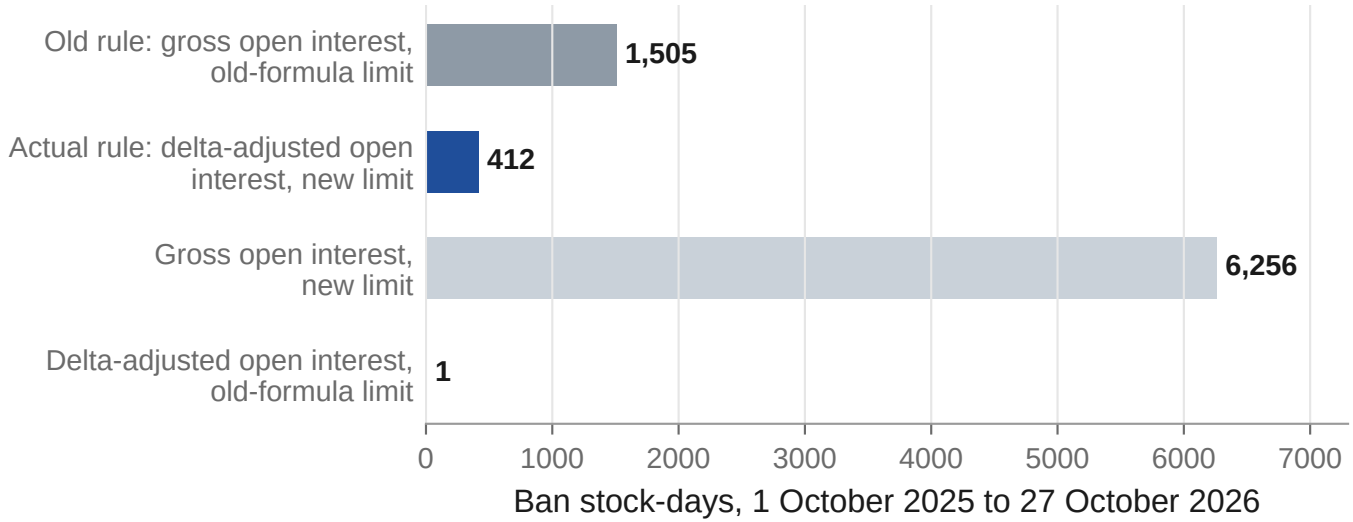


Figure 18: Ban stock-days, 1 October 2025 to 27 October 2026, under the actual and three counterfactual rules.

- Before 2025 bans came from rising open interest; only 0.4% from a falling limit.
- Delta-adjusted open interest removes almost all bans; the lower limit alone would have quadrupled them.

412 ban days against 1,505 under the old rule: about three quarters fewer.

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Entry into and exit from the F&O segment (H6)

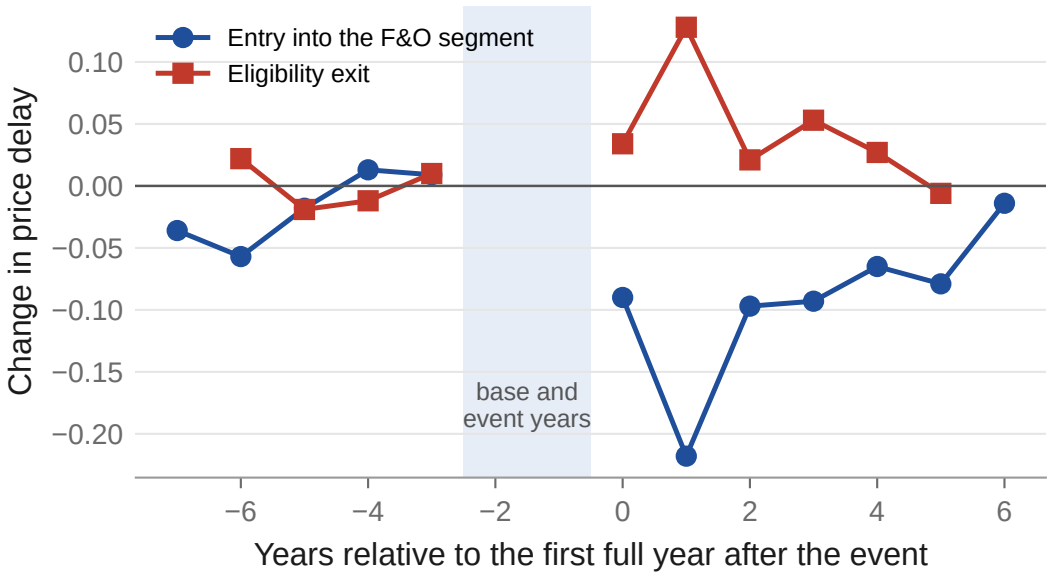


Figure 19: Change in price delay by year relative to the event.

Table 8: Pre-specified components

	Estimate	p
Entry: delay	$-\mathbf{0.109}$	10^{-6}
Entry: drift	-0.040	0.99
Exit: delay	$+0.053$	0.09
Exit: drift	$+0.014$	0.77

Sun and Abraham: -0.063 (s.e. 0.019). Selection on growth cannot be ruled out.

A constraint index for every listed stock

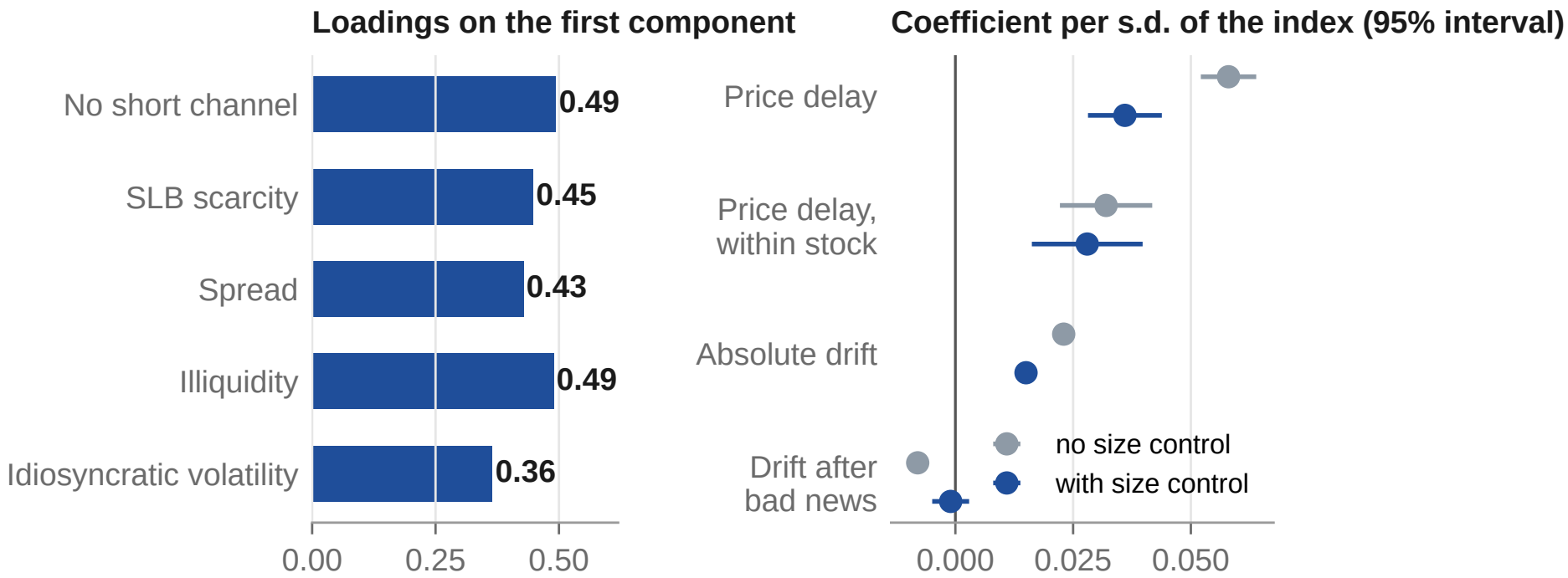


Figure 20: Loadings of five frictions on the first component (left); validation with and without a size control (right). 80,042 stock-quarters, 3,310 stocks.

- More constrained stocks adjust more slowly, also within stocks and after a size control.
- The index's link with drift after bad news disappears once size is controlled for.

Anomalies and short-sale constraints (H5)

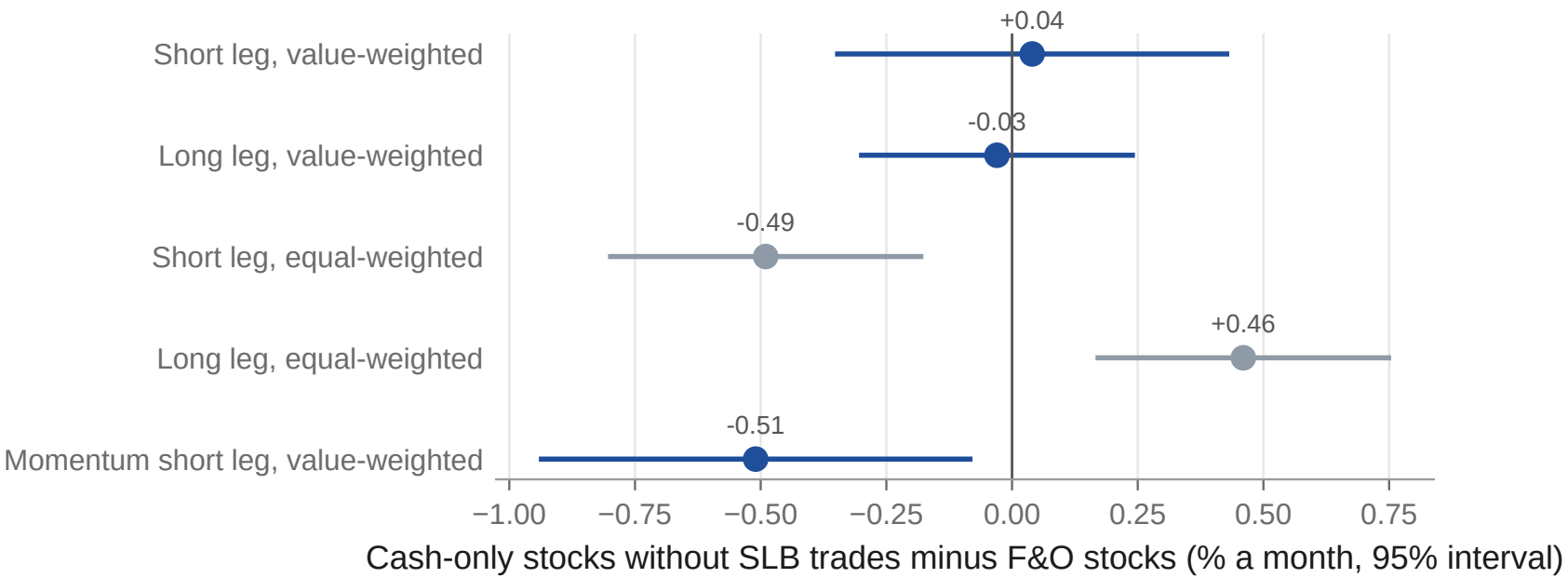


Figure 21: Cash-only stocks without lending minus F&O stocks, legs of a composite mispricing sort.

- Value-weighted, the overpriced side does no worse where shorting is hardest ($p = 0.57$).
- Equal-weighted, anomalies are stronger among small cash-only stocks, on both sides; momentum alone fits.

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Summary of the pre-specified tests

Hypothesis	Result	Comment
H8 trading	Supported	Identity holds under intraday checks
H3 costs	Supported	Weak against placebos
H6 short-sale access	Supported	Through price delay only
H7 one-sided correction	Rejected, opposite sign	Survives a noise correction
Level effect	$p = 0.083$	Signs as predicted
Density, RD at 95%	$p = 0.86$; -12.9 bp	Gap before the ban
Hysteresis	$p = 0.64$	No extra level shift
H9 anticipation, reform	$p = 0.019$; $p = 0.11$	Right direction
H2, H1, H4, H5	$p = 0.59,\ 0.002,\ 0.11,\ 0.57$	Mixed

Conclusions

- **Position limits** bind hard on trading; their price effect is real but not the predicted one-sided correction, and it fades within a week.
- **Enforcement** matters: the rule binds trade by trade only under intraday checks.
- **Short-sale access** speeds price adjustment, but evidence on overpricing and bad-news drift is mixed.
- **Trading costs** shape a wide, asymmetric band; its edges do not move one for one with the tax.
- **Arbitrage capital** leaves weak traces in the predicted direction.

Policy implications

What the evidence shows

- A ban cuts futures volume by almost 90%
- Its price effect appears at ban start and fades
- The 2025 rules cut ban days by three quarters

What it implies

- Bans restrain speculative demand at the cost of arbitrage
- Enforcement matters as much as the threshold
- Delta-based open interest removes avoidable bans

SEBI's study: 93% of more than one crore individual F&O traders lost money over FY22 to FY24 (SEBI, 2024a). The demand that position limits restrain is large.

Limitations and future scope

Limitations

- Daily data: noise inflates measured reversion
- Arbitrage inventory is not observed
- Only 44 ban spells since October 2025
- Size confounds cross-sectional comparisons

Future scope

- Intraday data for sharper tests of correction in bans
- A longer post-reform sample for the 2025 formula
- Extend the model for blocked new demand
- Observed-state smooth-transition models of the band

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Acknowledgement

I thank my supervisor, Prof. Aditi Chaubal, for her guidance throughout this project, and in particular for the advice to build a formal model of what the project would test before looking at any data.

Thank You

Questions and comments are welcome

Backup: the model's results and the tests

Result	Statement	Tested by
Band, comparative statics	Arbitrage only outside $[-c_-, c_+]$; costs set the edges	H3, H4
Inventory	The no-trade region moves with the existing book	Estimator tests
One-sided correction	In a ban, rich futures are not corrected	H7
Demand reversal, level	Blocked demand moves to cash; level sign not fixed	Level effect
Trading	Contract-level enforcement: $\Delta OI = -V$	H8
Anticipation, identification	Upper edge rises toward 95%; MWPL not an instrument	H9, density, RD
Short sales	Bad news slower without a short channel; Miller	H2, H5, H6

Backup: bans over time

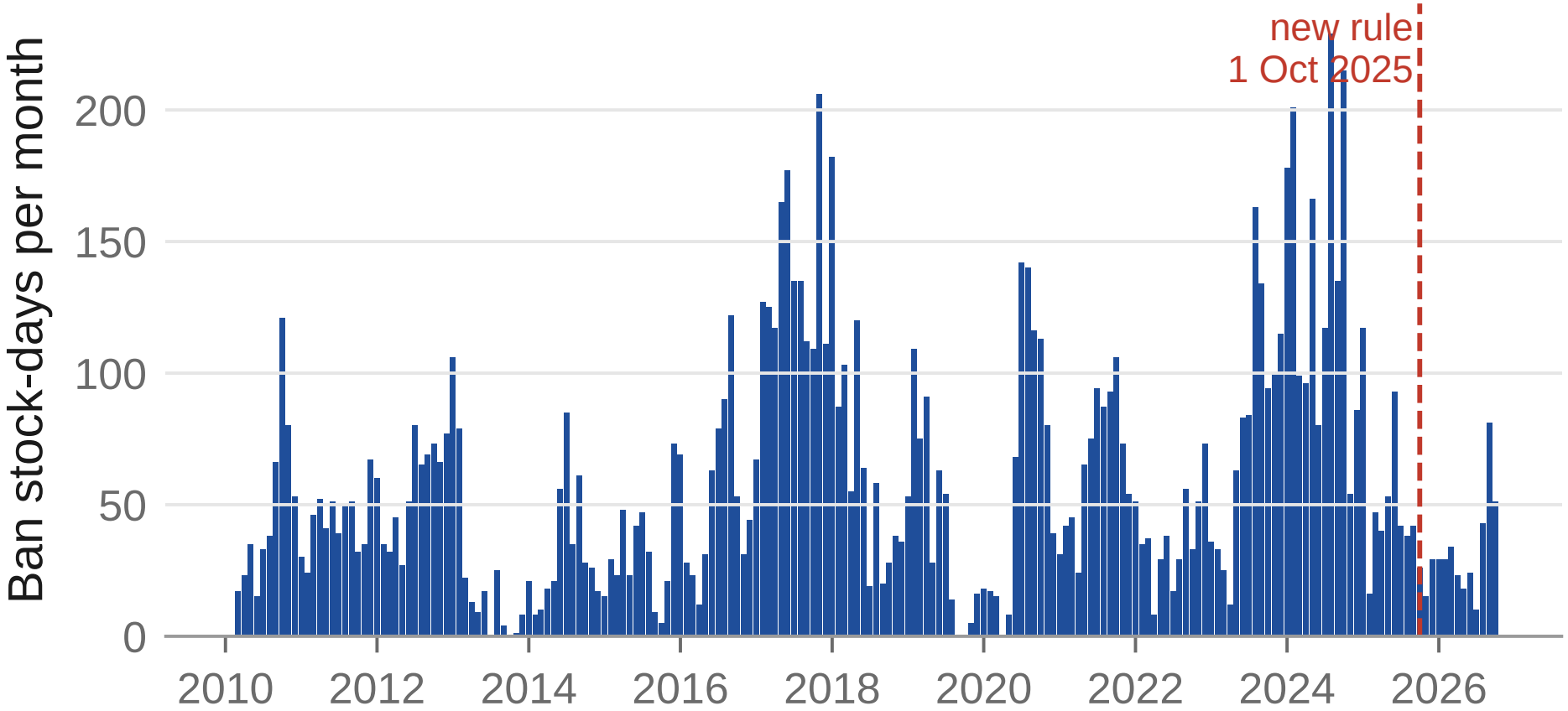


Figure 22: Stock-days in the ban per month, 2010 to 2026.

Backup: the October 2025 cut in the position limit

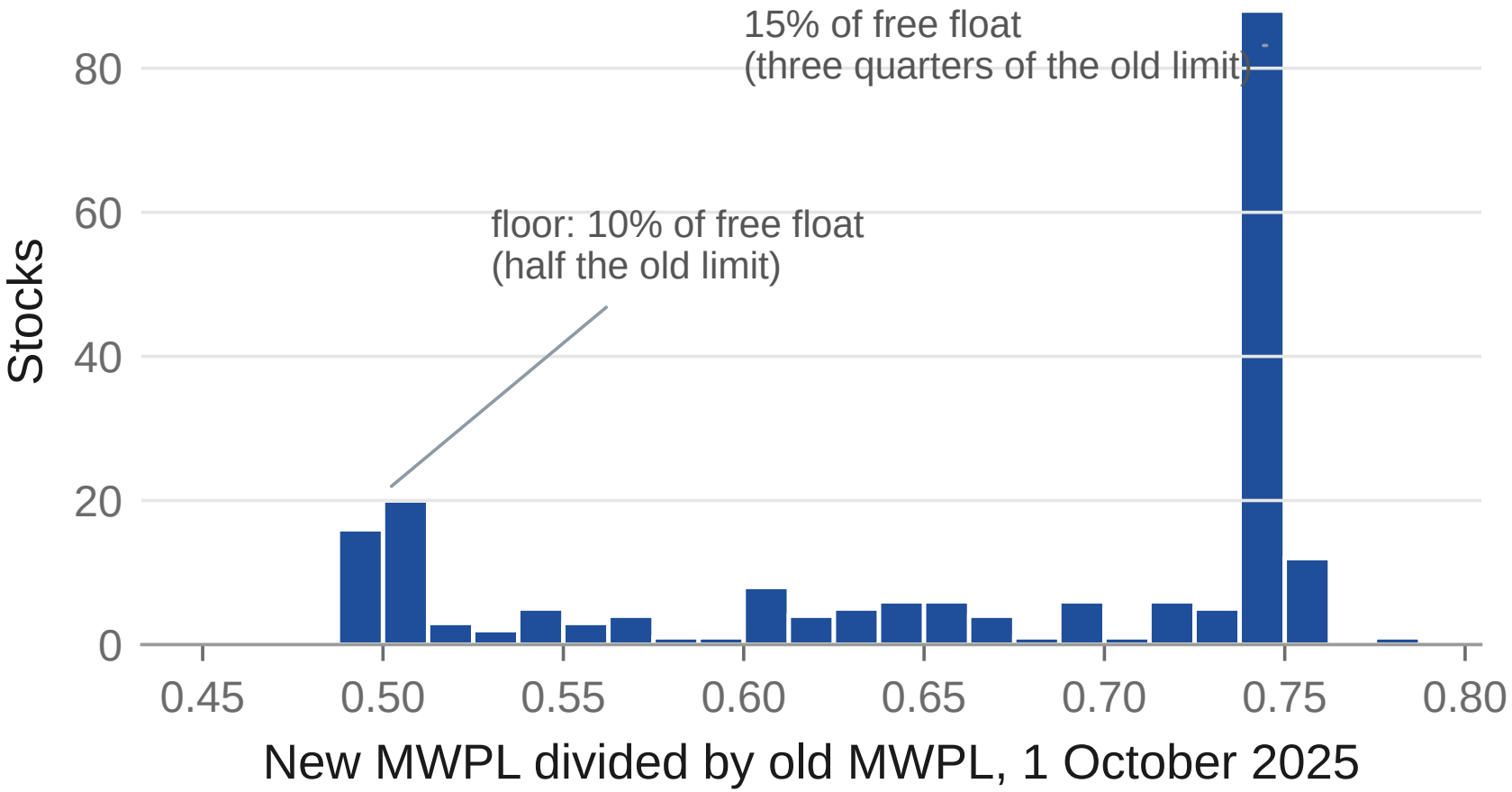


Figure 23: New MWPL divided by old MWPL, 1 October 2025 (208 stocks).

Backup: level effect by group

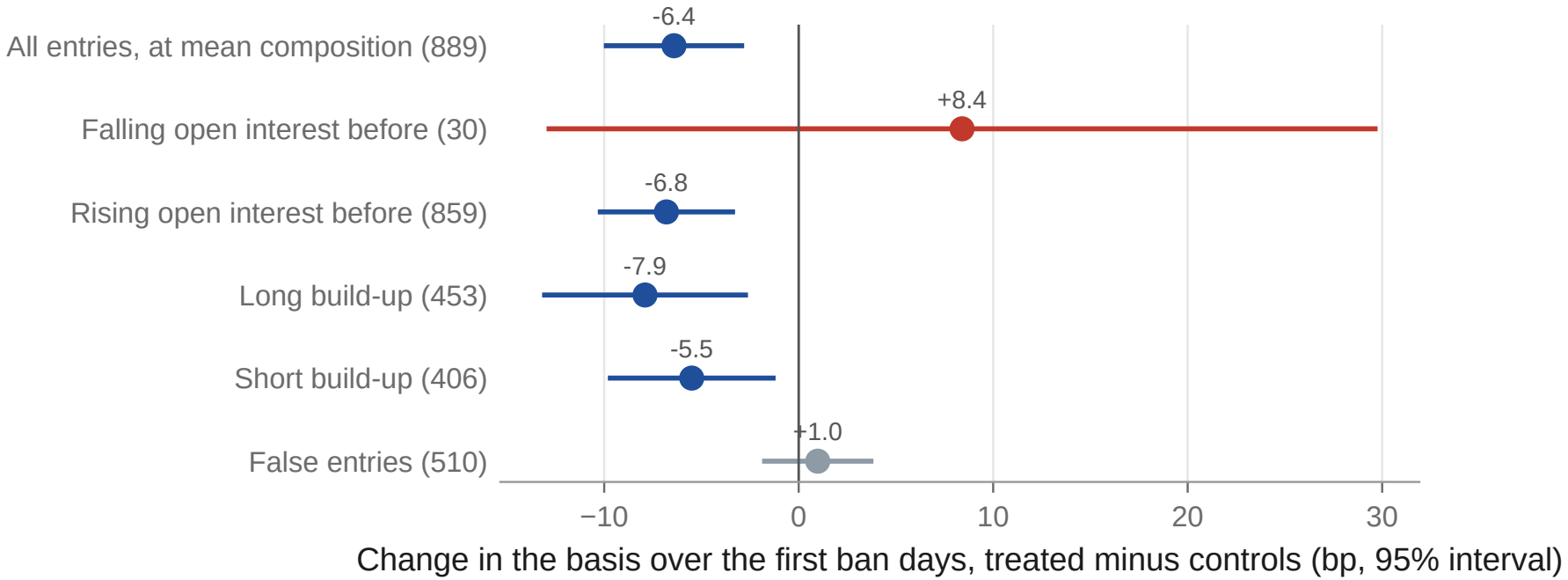


Figure 24: Change in the basis over the first ban days, treated minus controls, by composition and build-up type, with false entries.

Backup: the pre-analysis plan and its log

- Frozen with a git tag on 5 November 2026, after the supervisor's go-ahead and a full data audit; until then no outcome was related to any treatment.
- Every detail the plan left open (sample filters, composition splits, density population, outcome windows) was fixed in a dated log before the test that needed it.
- Analyses added after seeing a result are labelled exploratory; the pre-specified result is always reported first.
- Each estimator first recovered known effects on simulated data: the threshold model, the Callaway and Sant'Anna estimator and the matched level-effect design.